

DSA / Algorithms (Abdul Bari) - Day 21

45-day plan mapped to the Algorithms playlist

Dijkstra's Shortest Path

Single-source shortest path (non-negative weights).

Watch these Abdul Bari videos

> 3.6 Dijkstra Algorithm - Single Source Shortest Path - Greedy Method

Key concepts

- Greedy: expand the closest unvisited node.
- Relax edges to update distances.
- Priority queue $\rightarrow O(E \log V)$.

Key idea

```
dist[v] = min(dist[v], dist[u] + w(u, v)).
```

Tasks (1.5h)

- ☐ Watch the video
- ☐ Trace Dijkstra on a graph
- ☐ Implement it with a heap

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0